

LEARNING MECHANISMS IN TECHNOLOGY BUSINESS ACCELERATORS IN THE CONTEXT OF A NASCENT ECOSYSTEM

BACKGROUND

In recent decades, start-up business accelerators have proliferated across the world (Cohen, Fehder, et al., 2019; Hallen et al., 2020a). Well known Silicon Valley accelerators such as Y Combinator, and the many adopters that followed in its wake, emphasize learning as the primary mechanism by which ventures improve their performance (Hallen et al., 2020a). Indeed, Y Combinator was set up with the explicit goal of educating young technology entrepreneurs in business and finance (Cohen, Fehder, et al., 2019). Although some research on venture capital and organizational sponsorship suggests that non-learning mechanisms, such as sorting, (Mindruta et al., 2016) and signaling (Hallen, 2008) within accelerators may have a significant impact on venture performance, recent research provides compelling evidence of alternative learning mechanisms (Hallen et al., 2020a).

In a study of four accelerator cohorts from major metropolitan areas of the United States, Hallen *et al.* (2020a) distinguish experimentation learning, favored in most of the entrepreneurship literature, from broad, intense, and paced (BIP) consulting (consultative learning). Such trial-and-error learning can be validated scientifically by running frequent experiments that allow entrepreneurs to test elements of their business models, (Blank, 2003; Camuffo et al., 2020; Gavetti & Rivkin, 2007; McDonald & Eisenhardt, 2020; Osterwalder & Pigneur, 2010). Trial-and error learning is captured in tools and methods such as the lean startup methodology (Ries, 2011), the build-measure-learn feedback loop (Blank, 2003) and the business model canvas (Osterwalder & Pigneur, 2010).

Where trial-and-error learning is internally focused, emphasizing experimentation (repeated testing of hypotheses), BIP consultation is externally focused (interorganizational) and emphasizes knowledge transfer. In contrast to remote observation and imitation with the goal of gauging consumer response, consultative learning entails a closed loop of back-and-forth dialogues in which knowledge providers are actively involved. Although consultative learning appears to momentarily inhibit trial-and-error learning, it appears to expedite venture formation. This may occur when consultation prevents the entrepreneur from pursuing opportunities that should be avoided and saves the entrepreneur valuable resources (Cohen, Bingham, et al., 2019; Hallen et al., 2020a).

These insights rely principally on studies of prominent accelerators within developed entrepreneurial systems. Such systems are often rich in entrepreneurial knowledge, venture capital, a seasoned cadre of mentors, and knowledge of customer needs (Hallen et al., 2020a). The burgeoning research stream on business accelerators will benefit from studying venture learning within business accelerators in nascent entrepreneurial ecosystems for at least two reasons. First, learning mechanisms within accelerators in less munificent entrepreneurial ecosystems may differ substantially from those in developed systems, enabling development of alternative theories of venture learning. Second, the resource endowments and institutional conditions within nascent systems may enable scholars set boundary conditions on our emerging models of learning in accelerators.

We aim to address this gap in our study by asking the following questions: What type of learning happens in nascent system accelerators, and why? We address these questions in the context of start-up accelerators in Kazakhstan. We find that the learning mechanism in these accelerators is distinct; the principal learning mechanism in this nascent system is neither consultative type nor purely free-form experimentation based. We find that a hybrid learning mechanism, which we name *facilitated internal learning*, advocated, and enacted by the accelerators.

Our study makes two contributions. First, given the increasing prevalence of accelerators in the entrepreneurial landscape, and the rapidly-growing scholarship on early stage entrepreneurial learning (Cohen, Fehder, et al., 2019; Cohen, Bingham, et al., 2019; Hallen et al., 2020a) we provide evidence of a distinct learning mechanism within accelerators in nascent systems. As these systems mature such unique mechanism is likely to influence entrepreneurial practices and venture outcomes. Second, we question the generalizability of findings in extant literature regarding learning mechanisms in accelerators and by extension, early stage entrepreneurial learning (Cohen, Fehder, et al., 2019; Cohen, Bingham, et al., 2019; Hallen et al., 2020a). If we are to attend to a richer and complete understanding of early-stage entrepreneurial learning, the growing and evolving population of accelerators around the world, often in nascent systems, demands additional scrutiny. Our work is progress in that direction.

METHODS

Research Context: Kazakhstan's Nascent Technology Entrepreneurship Ecosystem

Kazakhstan has emerged as one of the most successful post-Soviet economies. Driven largely by its natural resource endowments and astute foreign policy, the country has achieved Upper Middle-Income status (World Bank Group, 2019). However, these gains in prosperity have outpaced institutional modernization. The competitive environment is still bedeviled by an outsized state-owned sector, weak regulatory institutions penetrated by vested interests and corruption (Bohr et al., 2019). Aiming to make a structural break from the past and place the country among the ranks of the 30 most developed countries by 2050, Kazakhstan's government has embarked on an ambitious program of digitalization, reformed enterprise laws, set up venture capital funds, and invested in technology parks.

The first acceleration programs in Kazakhstan were introduced in early 2010s by a small number of technology enthusiasts, who followed templates from accelerators in the West and from Singapore. By the late 2010s, as the government provided generous funds for acceleration programs, the number of accelerators in the country increased. Currently there are over 50 organizations within Kazakhstan that claim to be accelerators.

Even though Kazakhstan's accelerators attract increasing numbers of start-ups, the country still lacks the specialized intermediaries that are taken for granted in developed markets (Manikandan & Ramachandran, 2015). Venture capital firms are virtually nonexistent; credit rating agencies and insurance companies are in their infancy; formal contract enforcement may not inspire trust among entrepreneurs (Hoskisson et al., 2000); and there is few cadres of successful entrepreneurs to mentor new technology ventures. The nascency of the entrepreneurial ecosystem in Kazakhstan thus provides a valuable opportunity to investigate our research questions.

Data Collection

To address the research questions, we identify policymakers involved in supporting accelerators and all active accelerators in Kazakhstan. We interview managing directors to understand their program design elements and goals. In January 2021, we reduced our sample to three accelerators that were highly rated by our interviewees. Thereafter, between April 2021 and December 2022, we conducted interviews, performed participant observation and consulted archival documents.

Semi-structured interviews. We conducted 27 unstructured and semi-structured interviews with program sponsors, participants, program directors, and mentors to develop a deeper understanding of formal and informal learning mechanisms within these accelerators. In total we recorded over 20 hours of audio, on average interviews lasted 46 minutes.

Participant observation. We observed and surveyed two cohorts of participants from two accelerators (25 participants in total). In one accelerator, the program, lasting 12 weeks, was held online while the other program, held in-person, lasted eight weeks. In total we observed 27 hours of coaching and mentoring (tracking) sessions.

Documentary sources. We supplemented the interview and participant observation data with publicly available and internal archival data, such as participant cases, participant presentations, and graduate portfolios. We summarize the data collected in Table 1: Data Summary.

----- Insert Table 1: Data Summary around here -----

PRELIMINARY FINDINGS AND THEORIZATION

We find that that accelerators in Kazakhstan initially make a deliberate effort to adopt program design features like those of accelerators in mature systems such as Silicon Valley. Then, over a relatively short period of time, motivated by feedback from stakeholders and their own experience, accelerators modify program elements. These modifications are usually responses to specificities of the local context. We identify two features of the nascent environment which drives such modification: (1) small number and low diversity of knowledge sources; and (2) inexperience of participating ventures.

Crucially, these modifications in the imported program template shapes the learning mechanisms within these accelerators. While learning retains components of consultative or interorganizational learning, the emphasis is on internal learning. Moreover, to alleviate some of the pitfalls of internal

learning, accelerators track the ventures' experiments and trials and errors vigilantly, with the help of individual mentors called "trackers." Participant ventures consult frequently and systematically with trackers, who, in turn, map and facilitate venture progress.

Accelerators programs are characterized by hybridization wherein internal learning is prioritized with consultative learning mechanism *nested* within the former. Internal learning processes ultimately involve the creation of knowledge by individuals (Gavetti & Levinthal, 2000; Kessler et al., 2000), in contrast to interorganizational learning based on the knowledge and experience of other parties (Bingham & Davis, 2012; Lane & Lubatkin, 1998).

Accelerators facilitate creation of such individual knowledge, primarily using structured and mandatory consultation with trackers. A secondary implication of this mechanism is that learning becomes individually paced, unlike the intensive consultative learning that defines mature system accelerators (Hallen et al., 2020a). This carefully managed internal learning process is also more frugal than the broad consultative processes that we see in mature sources. We label this type of learning as "Facilitated Internal Learning" (FIL).

Distinguishing Hybrid Learning Mechanism in Nascent System Accelerators: Facilitated Internal Learning

In contrast to mentors in accelerators in mature systems, who mainly use their own experience during mentoring sessions and are free to use any approach, nascent system mentors follow a structured methodology. In Russia as well as in other Central Asian post-Soviet countries, this structure was first introduced by Internet Initiatives Development Fund (IIDF) in 2013 (IIDF). Responding to the lack of successful entrepreneur mentors in the region, the idea of a tracker was widely adopted by technology accelerators. Trackers are envisioned to be experts with some entrepreneurial experience, who are employed by accelerators throughout the duration of a program. As one tracker we interviewed put it, "trackers are mostly so called "downed pilots", experienced entrepreneurs whose ventures are currently in "cold" phase or haven't taken off" (Tracker #4, December 2022). Popular trackers will often make their services available to multiple accelerators, with program directors often sharing their contact list of trackers with each other.

Of the 13 program directors we interview, eight acknowledge that they extensively used tracker methodology, two mentions that they are using slightly modified version, while the remaining three accelerators design their own methods dynamically. The core set of concepts and routines in tracker methodology is illustrated in Figure 1.

In a generalized tracking system, all participants will typically set up weekly goals, fill a diary of trials and progress, report to and receive feedback from trackers in weekly traction meetings, and repeat this routine till the end of the program. Our analysis of progress diaries, observations of traction meetings and interviews with trackers provides a deeper insight on this process. Trackers themselves follow certain routines, such as scrupulously reviewing every venture's traction map and his own notes on the venture team's gaps and strengths on a weekly basis. Frequent consultations with the tracker are meant to be an exercise in joint assessment of the goals and their execution, as well as an exercise in proximate goal setting. As one tracker mentions: "the task of the tracker is to provide feedback and focus the team on the execution of the traction map." (Tracker #3, December 2022).

In principle, trackers are expected to provide feedback and direct mentees in their independent search for empirically validated solutions/answers. Such independent knowledge creation is the key element of internal learning (Ries, 2011). Broad, consultative interorganizational learning in mature system accelerators serve as a mechanism that forces search to remain open, so mitigating the frequent tendency among entrepreneurs to prematurely abandon the search for fresh ideas (Cohen, Bingham, et al., 2019). As reported in Table 2, our qualitative data indicates that the tracking methodology, through facilitated internal learning, plays the same role. By pacing the venture team to either speed up or slow down their search process, trackers make sure that ventures do not close their search prematurely.

-----Insert Table 2 here -----

Another salient aspect of such nascent-system acceleration programs is that they are frugal, in the sense that only those learning features deemed essential for a particular venture cohort are incorporated into the program design. Essential elements are chosen based on needs of the cohort (This

is because acceleration programs are often specialized in nature. For example, a program on financial technology will only induct fintech and associated ventures) and alumni feedback. Our interviews with accelerator sponsors and program directors reveal that a quick, efficient, and distraction-free search for a working business model is the primary premise that underlines all accelerator program design. An important implication of this frugal design is that consultation-related learning from external or embedded knowledge sources do not exceed 10 hours per week in these programs, are carefully curated and paced throughout the duration of a program. This stands in sharp contrast to the up to 40 hours per week of broad and intensive consultation-based activities that one finds in prominent accelerators in the West (Hallen et al., 2020a). We find that participants make up this slack time through activities like experimentation, trial and error and discovery driven planning, all key elements of internal learning (Bingham & Eisenhardt, 2011; McGrath & MacMillan, 1995). We illustrate comparison of program intensity between nascent and mature system accelerators in Table 3.

-----Insert Table 3 here-----

DISCUSSION

Mature system accelerators use specific program designs to address and mitigate bounded rationality challenges that beset ventures and entrepreneurs (Cohen, Bingham, et al., 2019; Grimes, 2018). These program elements are geared towards increasing the amount of external information searched by the entrepreneur, often by reigniting search for problems that they already believe to be solved. This is achieved through enabling broad, intensive, and paced consultation with a wide array of knowledge sources (Hallen et al., 2020a).

Learning in mature system accelerators thus presupposes multiple embedded knowledge sources as well as a wider variety of knowledge sources than peer networks (Cai and Szeidl, 2018; Hallen et al., 2020). Such multiplicity and diversity of knowledge sources in the form of domain experts and entrepreneur-mentors is scarce in nascent systems. For a nascent system accelerator trying to follow the mature system template, this resource constraint is all too salient. Accelerators in this system thus responds by reducing or eliminating processes that require domain-expert consultations. In its place, a new system emerges where mentors play a larger and more salient role in the acceleration process, in the form of tracking. This new system generates a learning mechanism that we label *facilitated internal learning* (FIL), which exhibits features of experimentation and trial and error learning, but which is structured and facilitated by the accelerator in the form of tracking. Bingham and Eisenhardt (2011) studied trial-and-error learning, in which entrepreneurs combine learning and action in real time to identify more effective strategies. Others study experimentation wherein firms reduce uncertainty and move ahead with fewer mistakes by engaging in deliberate offline learning. Experimentation plays a prominent role in product development processes (Eisenhardt & Tabrizi, 1995; Thomke, 1998) and in the customer exploration efforts of “lean” startups (Bremner & Eisenhardt, 2019; Ries, 2011). Nascent system accelerators thus address the bounded rationality challenges of new ventures with a different organizational and program design, that advances a distinct learning mechanism separate from mature system counterparts.

The relative lack of prior entrepreneurial experience of venture founding teams is another distinguishing feature of nascent system accelerators. The proportion of experienced entrepreneurs with substantial pre-entry knowledge (Dencker et al., 2009; Gruber et al., 2013) is extremely sparse in cohorts that we observe in nascent system accelerators. In mature system contexts where this will not be the case to the same extent, experienced entrepreneurs may benefit from additional, consultative learning, such as that provided by accelerators, because numerous entrepreneurial opportunities include a significant level of novelty. Accordingly, it may be that FIL in nascent system accelerators is driven by the logic of learning what to do in the form of local search and limited deliberation around planned activities. (Gavetti & Rivkin, 2007; McGrath & MacMillan, 1995).

CONCLUSIONS

Extant literature emphasizes learning as the primary mechanism by which ventures improve their performance (Hallen et al., 2020a). These insights are primarily based on studies of notable accelerators within established entrepreneurial systems. Such systems typically contain a wealth of entrepreneurial

knowledge, venture money, mentorship opportunities, and customer-needs information (Hallen et al., 2020b) . It is not yet clear that these findings apply to less developed entrepreneurial systems. Accelerators in less developed entrepreneurial ecosystems may have different learning mechanisms than those in developed systems, allowing richer theorization of venture learning. Moreover, the resource and institutional conditions within nascent systems may enable scholars set boundary conditions on our emerging models of learning in accelerators.

Our experience with prominent accelerators in Central Asia, one such nascent entrepreneurial ecosystem, indicates that the learning mechanism in accelerators is distinct. Preliminary analysis reveals that the learning mechanism in nascent system accelerators is neither consultative type, nor is it purely free form experimentation based. These are the two most favored learning mechanism in the literature on learning in ventures. Instead, we find that a hybrid learning mechanism, which we name *facilitated internal learning*, is being advocated and enacted by the accelerators. This mechanism emphasizes trial and error but track the experimentation process diligently through frequent consultation with trackers. We then put forward two context specific factors that may have led to the emergence of this hybrid form.

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LIST OF FIGURES AND TABLES

Table 1: Data Summary

Interviews		27
	Program sponsors	5
	Program directors	13
	Trackers (Startup Mentors)	5
	Participants	3
	Investor	1
Observations		12 weeks
	Pre-seed acceleration program (online)	12 weeks
	Pre-seed acceleration program (in-person)	8 weeks
Archival documents		29
	Program designs	2
	Presentations of school of trackers	8
	Workbook school of trackers	3
	Participant diaries	4
	Participant presentations	11
	Graduate portfolios	4

Table 2: Facilitated Internal Learning

Characteristic	Definition	Illustration
Frugality	Learning elements that are deemed essential are those that gets incorporated in program design	“We screen startups after we select them, understand what is needed and what is not, and develop the program”: Program director of accelerator B “Unnecessary workshops and consultations can confuse startups. Experts are needed exclusively upon request, except basics”: Tracker 2, mentored over 100 startups
Individually paced	Participants’ short and intermediate goals are paced and tracked in a tailored manner.	“First participants set their goals and then the measurements are made whether the participant has achieved their weekly goals or not. If not, then they keep working on it, if yes, then they set new goals”: Program director of accelerator A “The tracker sets the pace, sometimes I speed them up, sometimes I slow them down”: Tracker 5, mentored 10 startups
Internal learning	Emphasis on independent knowledge creation by individuals, mostly through trial-and-error.	“In between traction meetings, participants are expected to execute their weekly plans: test hypotheses, hold meetings, send commercial offers, search for a potential customer and record outcomes in their progress diaries”: Program director of accelerator C. “The participants independently execute the assigned weekly tasks in accordance with the milestones indicated at the beginning in the traction map”: Tracker 4, mentored over 500 startups

Table 3: Program intensity

Criteria	Acceleration programs in a mature ecosystem	Acceleration programs in a nascent ecosystem
Number of different knowledge sources involved in the program	50 to 150	20 to 30
Individual meetings with different mentors during the program	40+	>10
Consultation-related learning (hours per week)	40+	3-10

Figure 1. Tracking Cycle